

SHARYPOV, Russian Federation

WMO: 295580

Lat: 55.5344N

Lon: 89.2053E

Elev: 1046

StdP: 14.15

Time Zone: 7.00 (E07)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.3	-25.2	-35.6	0.8	-31.2	-30.0	1.1	-25.0	22.2	15.1	18.5	17.8	1.6	160	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	83.4	65.3	80.5	64.6	77.6	63.1	68.3	79.3	66.8	76.9	65.2	74.4	5.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.6	94.8	73.4	62.9	89.1	71.5	61.4	84.6	70.2	33.3	78.8	32.1	77.2	30.8	74.2	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	16.7	14.5	13.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)				N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.3	3.9	7.5	23.7	38.9	47.6	62.6	64.9	61.4	49.2	35.3	16.3	11.0
	DBStd	24.07	14.75	14.68	11.52	9.64	8.95	7.70	4.98	5.94	8.08	9.62	14.88	15.33
	HDD50	6747	1429	1190	815	348	159	11	0	1	110	464	1012	1208
	HDD65	10976	1894	1610	1280	784	540	129	64	143	476	921	1462	1673
	CDD50	1396	0	0	1	13	85	387	461	355	87	7	0	0
	CDD65	151	0	0	0	0	2	56	59	32	2	0	0	0
	CDH74	1547	0	0	0	3	46	666	511	276	45	0	0	0
	CDH80	302	0	0	0	0	2	158	92	43	7	0	0	0
Wind	WSAvg	5.5	5.7	5.3	5.9	6.8	6.6	4.8	4.2	4.3	5.0	5.8	6.2	6.0
Precipitation	PrecAvg	21.2	0.8	0.7	0.9	1.3	1.8	2.8	2.9	3.0	2.1	1.8	1.6	1.3
	PrecMax	24.7	1.7	1.5	2.2	2.9	3.5	5.1	4.6	5.9	4.4	3.1	2.6	2.4
	PrecMin	17.2	0.1	0.2	0.2	0.2	0.7	1.6	1.2	1.1	0.3	0.9	0.8	0.4
	PrecStd	2.2	0.4	0.3	0.5	0.6	0.8	0.8	0.9	1.1	0.9	0.6	0.5	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.1	38.8	53.9	71.9	78.8	88.5	86.6	84.4	80.0	66.1	50.4	38.3
		MCWB	31.3	33.3	43.5	50.7	55.7	66.1	68.5	68.6	59.3	50.1	44.1	33.6
	2%	DB	32.7	34.3	47.2	63.4	74.2	84.2	82.2	80.5	72.9	58.5	41.8	34.9
		MCWB	29.0	30.1	37.8	47.0	54.2	64.0	66.8	65.6	58.9	47.1	36.6	31.4
	5%	DB	28.3	31.6	42.9	59.5	69.7	81.1	79.5	76.9	67.0	53.2	37.5	32.3
		MCWB	25.1	28.2	36.0	45.8	52.1	63.5	65.7	64.3	55.7	45.3	33.1	29.1
	10%	DB	24.6	28.4	39.2	54.5	64.1	77.9	76.5	73.6	62.7	48.6	34.1	29.4
		MCWB	21.8	25.4	33.3	43.5	50.5	62.1	64.5	63.1	53.6	42.4	30.8	26.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.1	34.0	43.2	52.1	58.8	70.0	70.9	70.5	62.0	51.6	43.5	34.1
		MCDB	35.6	38.3	52.0	65.7	72.7	82.7	83.6	80.4	74.7	62.3	49.9	37.2
	2%	WB	28.6	30.8	39.1	49.6	56.3	67.7	68.5	67.3	59.2	48.6	36.5	31.9
		MCDB	32.6	33.7	45.9	60.9	68.5	78.5	79.4	78.0	69.4	57.3	40.5	34.5
	5%	WB	24.9	28.2	36.2	46.8	54.1	65.8	67.1	65.6	57.1	45.8	33.3	29.2
		MCDB	28.1	31.4	42.3	57.1	66.7	76.6	77.2	74.5	66.5	52.5	36.6	32.0
	10%	WB	21.5	25.0	33.7	43.9	51.7	63.9	65.5	63.9	54.5	42.8	31.0	26.4
		MCDB	24.5	28.2	39.2	54.1	63.8	74.7	74.3	71.8	61.4	49.0	34.3	29.7
Mean Daily Temperature Range	5% DB	MDBR	15.1	15.8	16.9	18.4	19.4	21.3	18.0	18.1	17.0	13.5	12.5	13.3
		MCDBR	15.2	15.2	18.9	26.4	28.9	27.0	24.2	24.0	25.8	21.8	12.8	13.0
		MCWBR	12.7	12.3	12.4	13.6	13.8	11.1	10.9	11.6	12.7	12.6	9.2	11.1
	5% WB	MCDBR	15.4	13.9	18.0	23.2	24.1	22.4	20.6	20.7	22.9	20.1	12.6	13.1
		MCWBR	13.1	11.4	12.4	13.7	12.7	10.5	10.0	10.6	11.8	12.0	9.4	11.4
Clear-Sky Solar Irradiance	taub	0.272	0.307	0.354	0.393	0.403	0.400	0.423	0.405	0.365	0.343	0.305	0.273	
	taud	2.050	2.030	1.970	2.119	2.209	2.311	2.248	2.272	2.333	2.243	2.064	1.987	
	Ebn at Noon	195	233	249	260	266	268	258	255	250	221	176	158	
	Edh at Noon	21	31	43	43	42	38	40	36	30	25	21	17	
All-Sky Solar Radiation	RadAvg	266	577	1027	1423	1575	1839	1644	1344	895	476	238	166	
	RadStd	28	49	75	84	177	173	114	118	104	47	32	20	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page